

Student Data Analysis Project

Figure 1

]:						
	Student_ID	Math	Physics	Chemistry	English	Computer_Science
0	S1	78	41	44	91	87
1	S2	91	91	58	59	75
2	S3	68	70	62	76	60
3	S4	54	79	56	95	84
4	S5	82	54	65	55	57

Figure 2

```
print(data.head())
```

	Math	Physics	Chemistry	English	Computer_Science
0	78	41	44	91	87
1	91	91	58	59	75
2	68	70	62	76	60
3	54	79	56	95	84
4	82	54	65	55	57

Figure 3

```
vectors = data.values
print("First student vector:", vectors[0])
First student vector: [78 41 44 91 87]
```

```
v1 = vectors[0]

norm1 = norm(v1, 1)
norm2 = norm(v1, 2)

print("L1 Norm:", norm1)
print("L2 Norm:", norm2)

L1 Norm: 341.0
L2 Norm: 159.84680165708664
```

```
v2 = vectors[1]

dot = np.dot(v1, v2)

angle = np.arccos(dot / (norm(v1)*norm(v2)))

print("Dot Product:", dot)
print("Angle (radians):", angle)

Dot Product: 25275
Angle (radians): 0.381783498557253
```

```
cross = np.cross(v1[:3], v2[:3])
print("Cross Product:", cross)

Cross Product: [-1626 -520 3367]
```

Figure 4

```
Projection of v1 on v2: [79.22378754 79.22378754 50.49428217 51.36487324 65.29433039]
```

```
matrix = data.values  
print("Matrix shape:", matrix.shape)
```

```
Matrix shape: (500, 5)
```

```
transpose = matrix.T
```

```
product = np.dot(matrix, transpose)
```

```
# take small square matrix  
square_matrix = matrix[:3, :3]  
  
print("Determinant:", det(square_matrix))
```

```
Determinant: 61785.99999999991
```

```
inv_matrix = inv(square_matrix)  
print("Inverse:\n", inv_matrix)
```

```
Inverse:  
[[ 0.02560451  0.00870747 -0.02631664]  
 [-0.02748195  0.02984495 -0.00841615]  
 [ 0.00294565 -0.04324604  0.05449455]]
```

```
print("2D = line, 3D = plane, higher = hyperplane")
```

```
2D = line 3D = plane higher = hyperplane
```

Figure 5

```
print("\nEigenvalues: ", eigenvalues)  
Eigenvalues: [201.71996635 253.62241501 299.54094905 406.94466207 371.3598031 ]
```

```
P, L, U = lu(square_matrix)
```

```
print("L:\n", L)  
print("U:\n", U)
```

```
L:  
[[ 1.          0.          0.          ]  
 [ 0.85714286  1.          0.          ]  
 [ 0.74725275 -0.05405405  1.          ]]  
U:  
[[ 91.          91.          58.          ]  
 [ 0.          -37.          -5.71428571]  
 [ 0.           0.          18.35046035]]
```

```
U_svd, S, Vt = svd(matrix)
```

```
print("Singular Values:", S)
```

```
Singular Values: [3495.2644025  450.56011123  409.17360746  375.19373163  330.83525428]
```

Figure 6

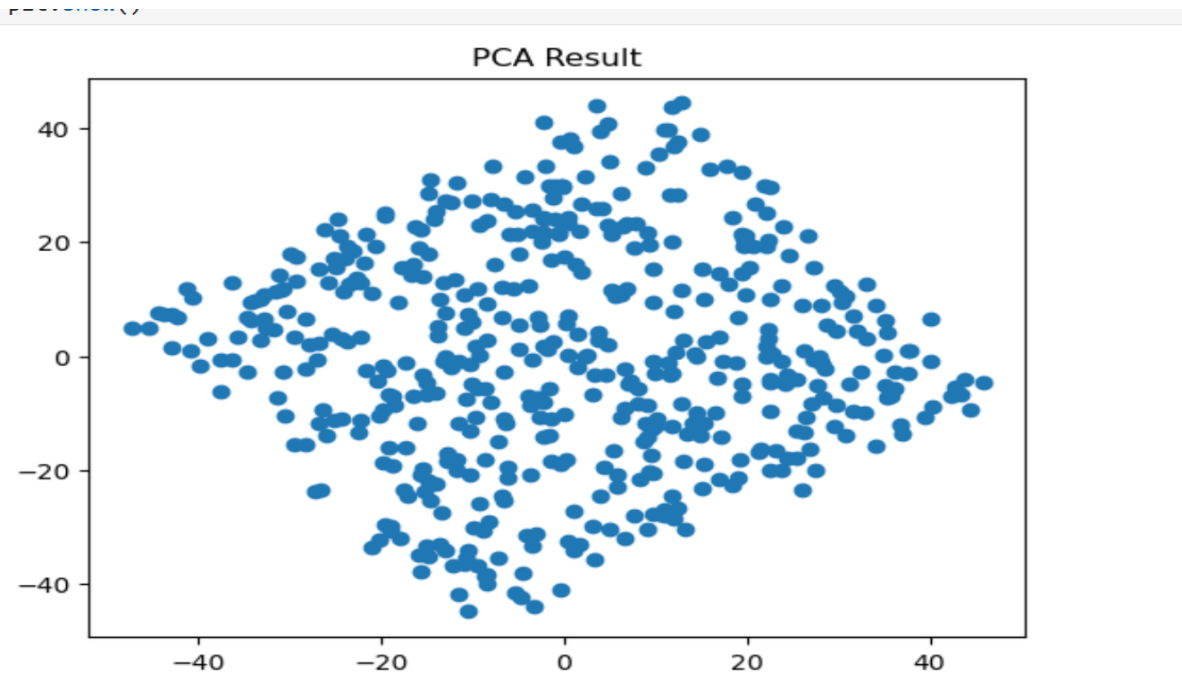


Figure 7

